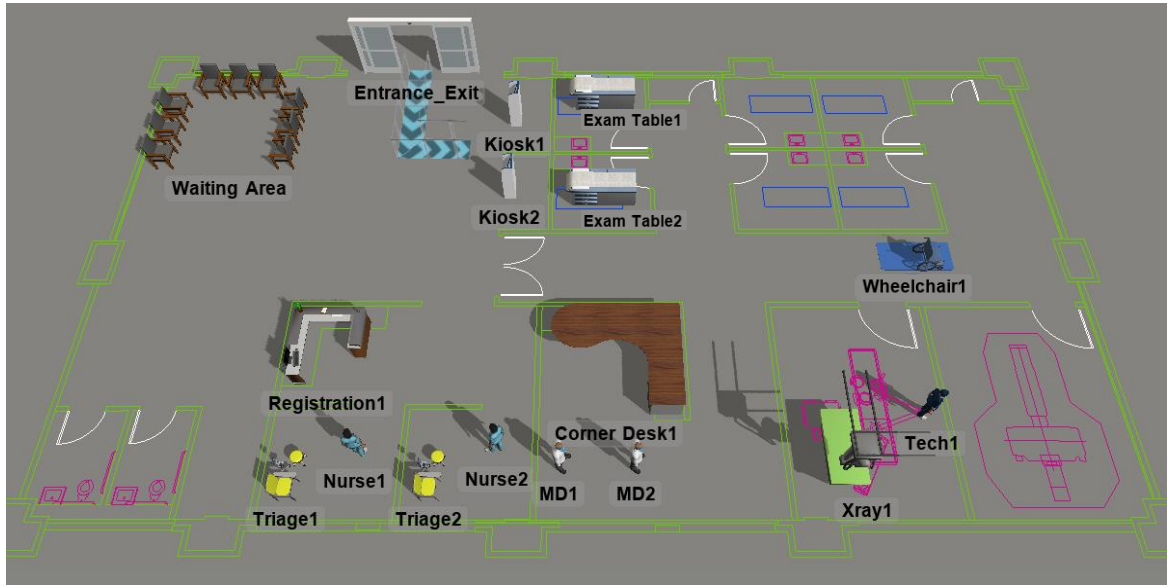
FlexSim Healthcare Modeling Guide

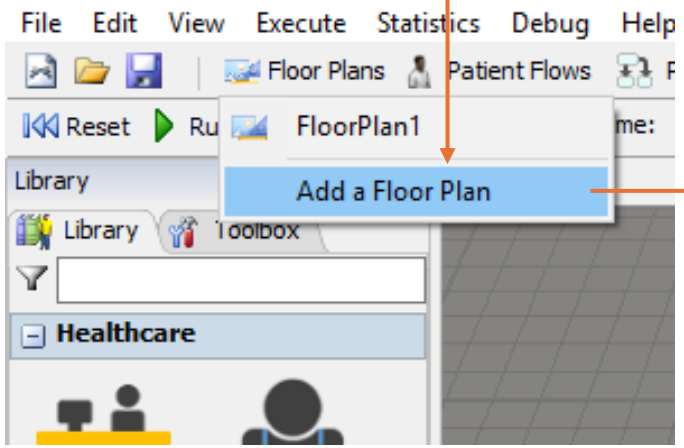
Part 1: Build 3D Model

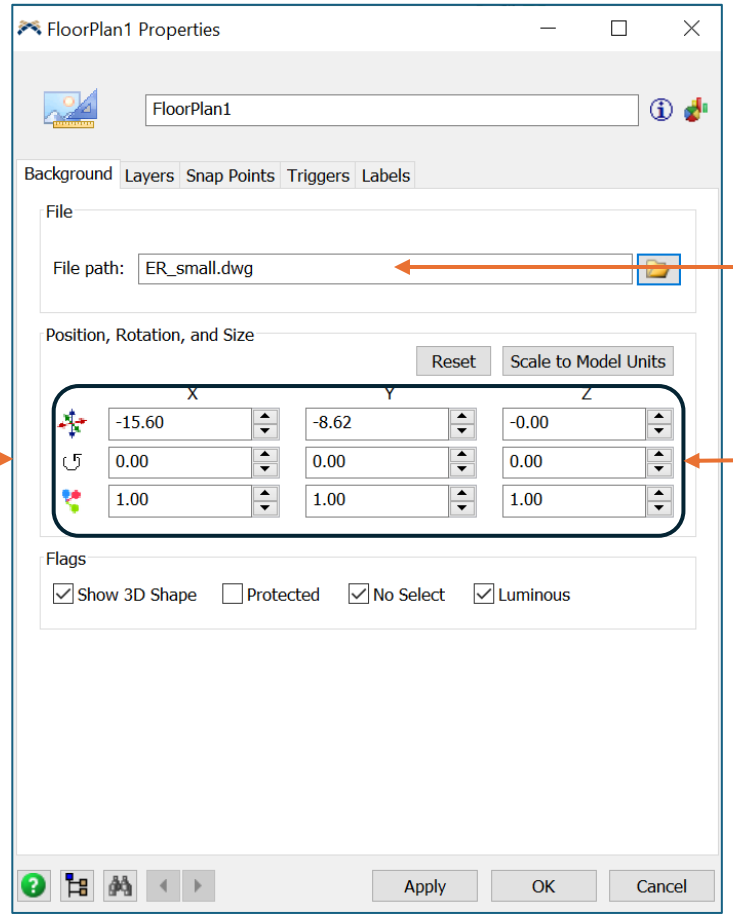
- Add floor plan
- Add Location objects
  - Kiosk, Registration, Triage, Exam Table, Xray
- Add Prop
  - Glass Open door, Corner desk
- MultiLocation
  - Chairs
- Add Waiting Line
- Add Staff
  - MD, RN, Tech
- Add Transport
  - Wheelchair
- Rotate & Resize Objects
- Create groups



Add Floor Plan

1. Find Floor Plans at the Toolbar. Click Add a Floor Plan.





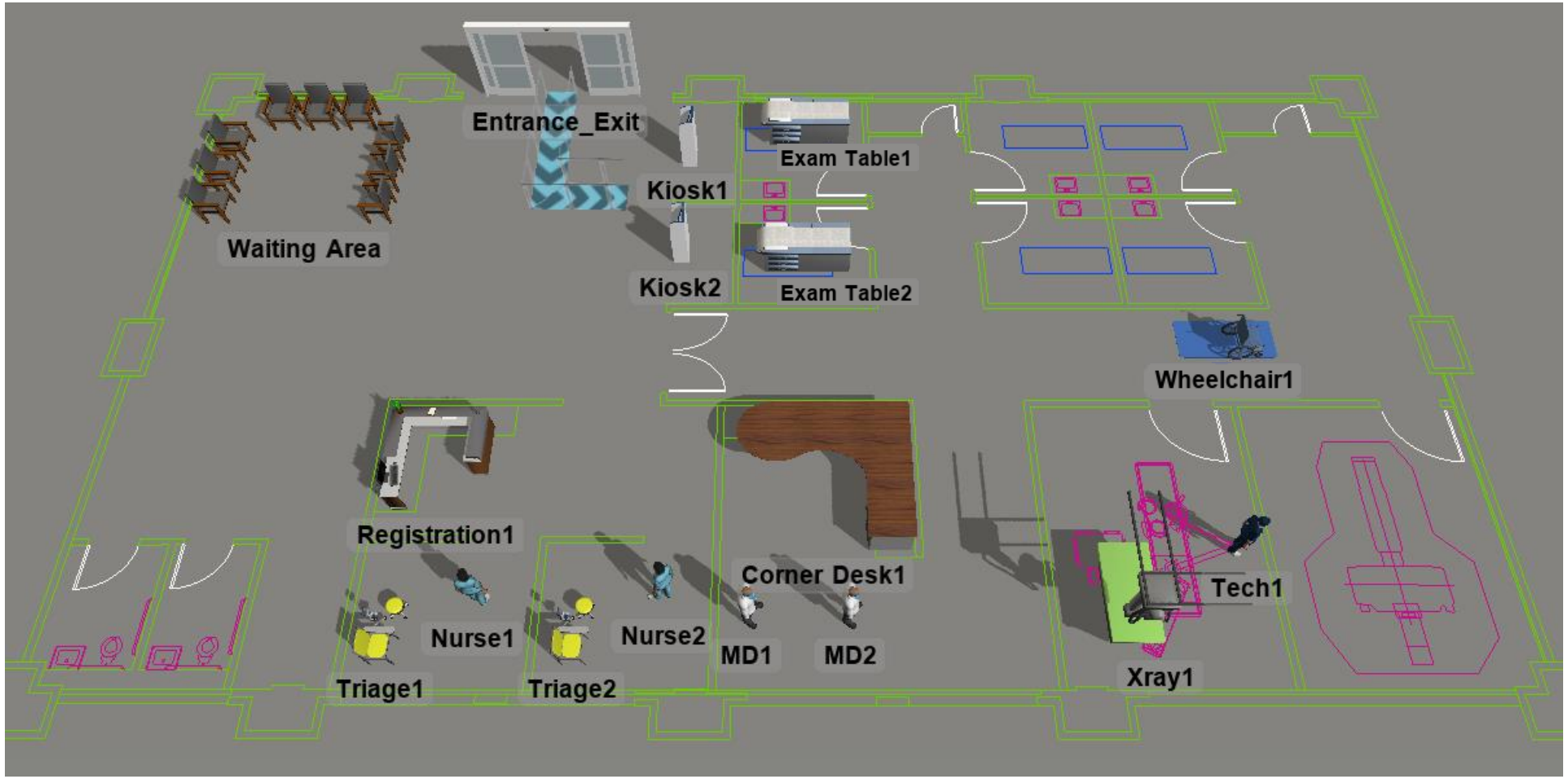
2. Select the AutoCAD file ‘ER_small.dwg’ in the file path.

3. Verify the values.

Add 3D Objects

- Under Healthcare group, except for Waiting Line and Elevator Bank, clicking on any icon will open a menu of individual objects.
- Click and drag an object (such as Kiosk) into the 3D model.
- Create the required 3D objects this way and place them accordingly (refer to next slide).

Layout of the Clinic

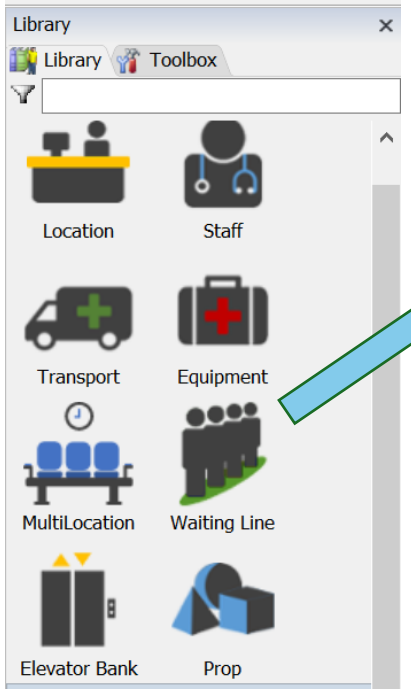


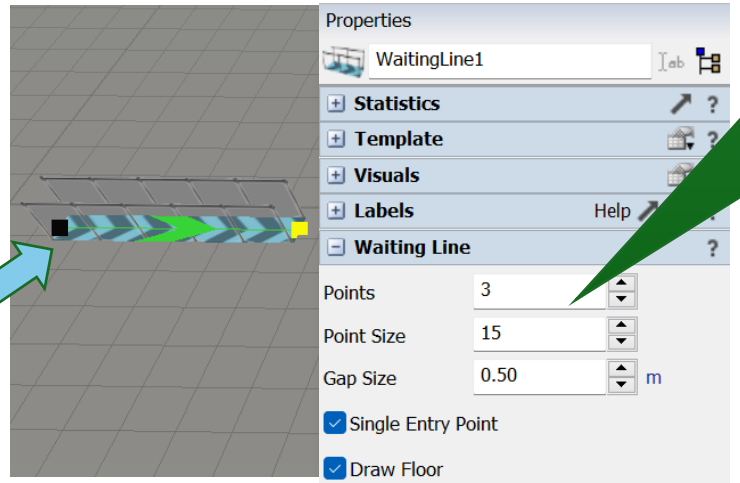
Follow the layout as shown to build the 3D model.

Rename the objects as necessary.

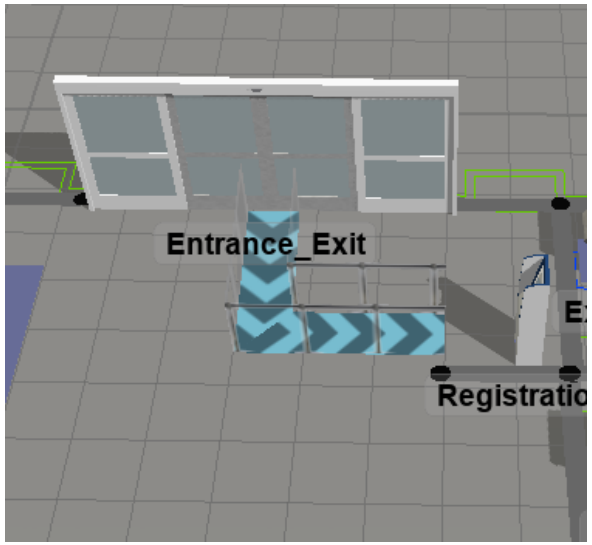
The names of the objects will be referenced later on.

Adjust Waiting Line

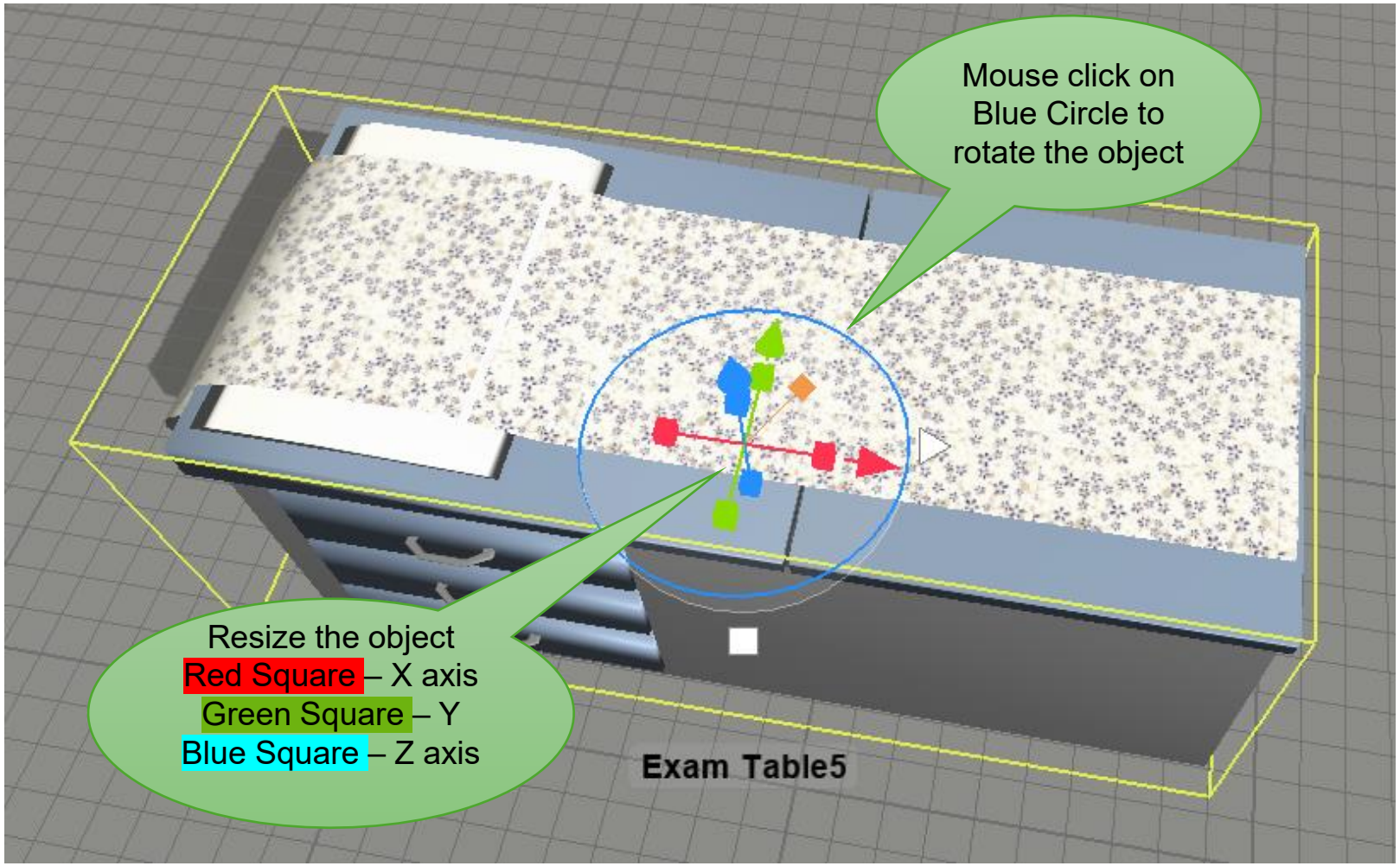




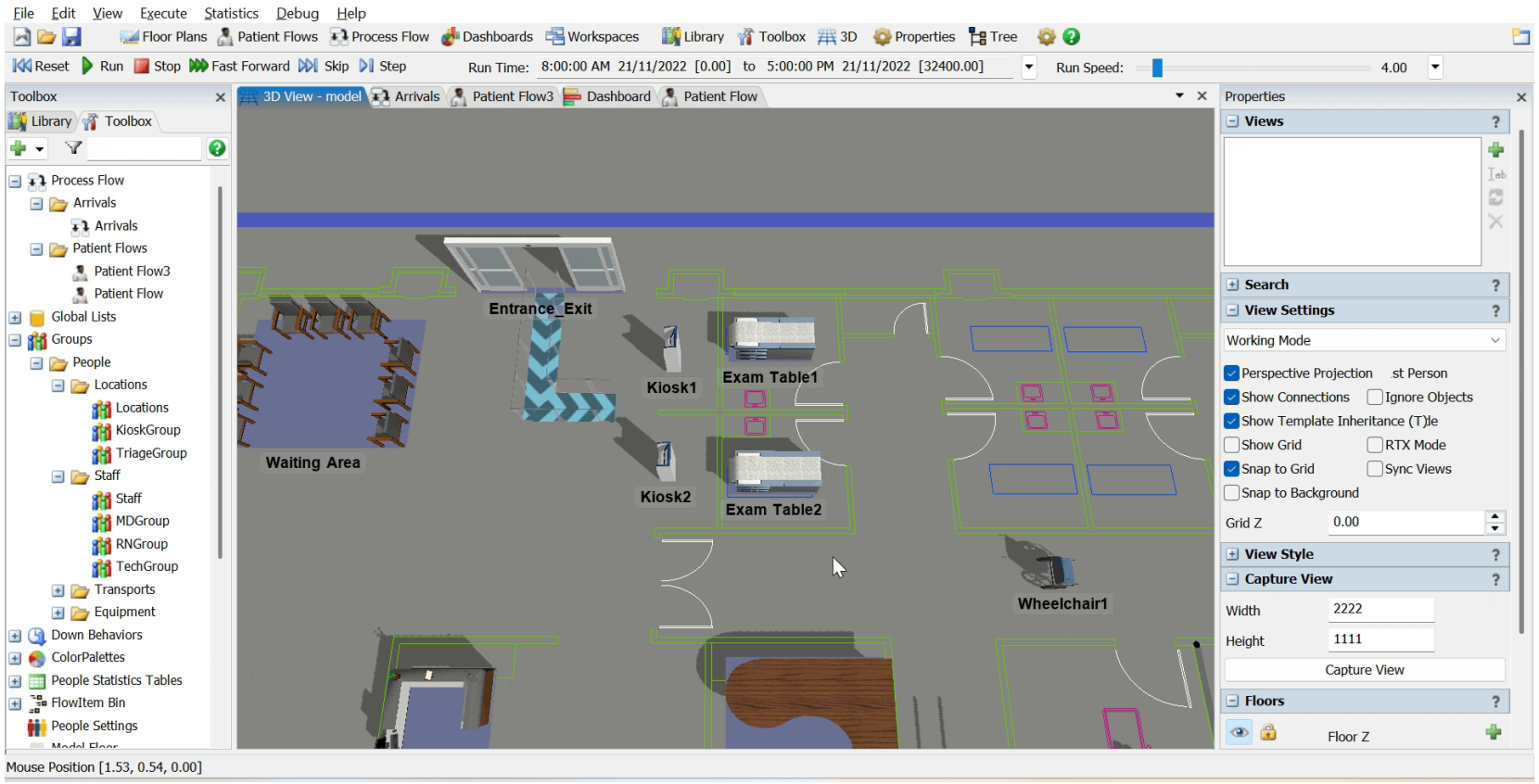
Edit points to adjust waiting line shape.
In the Properties, change Points from 2 to 3 and the Gap Size to 0.5 m to so that there is a 0.5 meter separation between people.



Rotate & Resize Objects



Create Location Groups



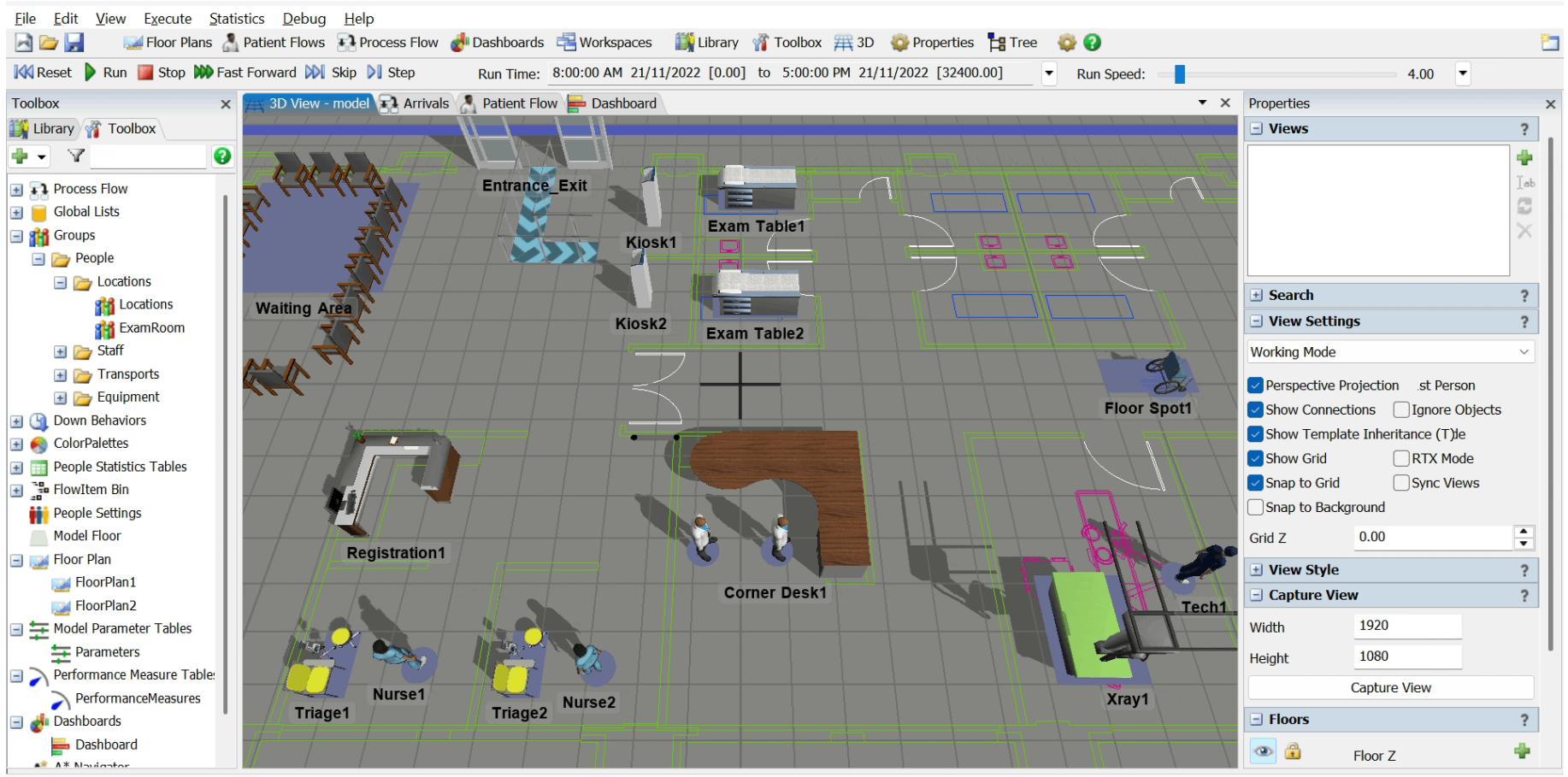
1. Press and hold down the Shift key while using your mouse to draw a box around the two **Exam Table** objects in your model. A red box will appear around the Exam Tables to show they are selected. Alternatively, hold down CTRL key to select.

2. Right-click **ExamTable1** to open a menu. Point to **Location Groups**, then select **Add to New Group** to open a group properties window.
Rename the group **ExamRoom**.
Notice that the Two **Exam Tables** have become members of the group.

3. Press and hold down the Shift key while clicking a blank area in the 3D model to de-select the Exam Tables.

4. Repeat the steps above to create a **KioskGroup** and a **TriageGroup.**

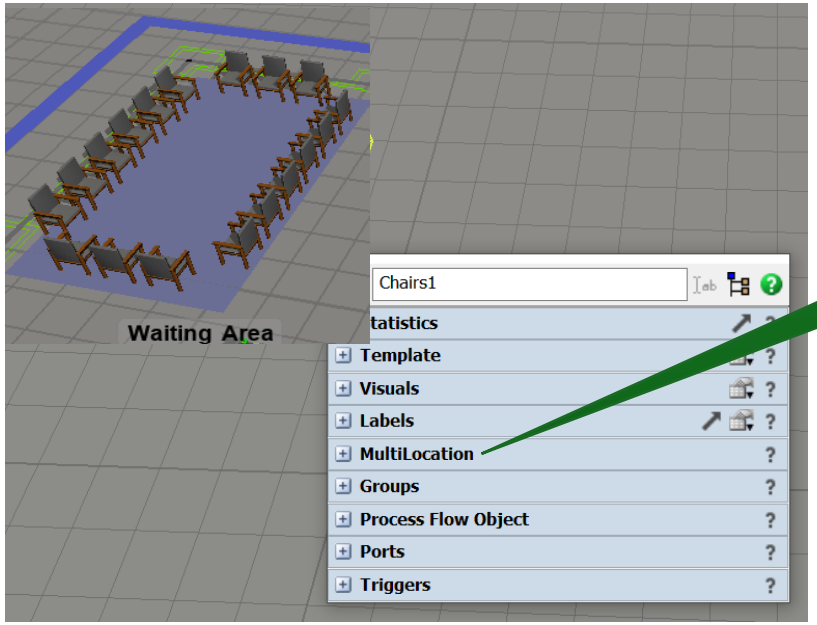
Create Staff Groups



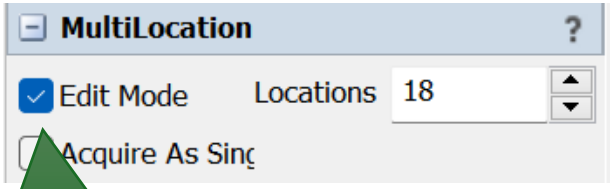
Under Staff Groups, create

1) A nurse group (RNGroup)
2) A doctor group (MDGroup)
3) A Tech group (TechGroup)

Add More Chairs in Waiting Area



The MultiLocation panel allows you to edit the number and placement of individual chairs.



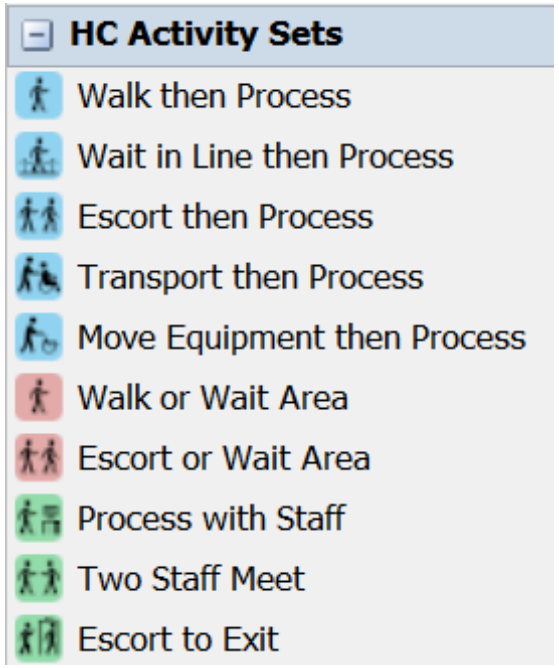
Edit Mode
Checking this box allows you to move individual chairs
Locations
Set the number of chairs = 18
After aligning the added chairs, uncheck the Edit Mode box.

REPUBLIC POLYTECHNIC

Part 2: Create Patient Flow

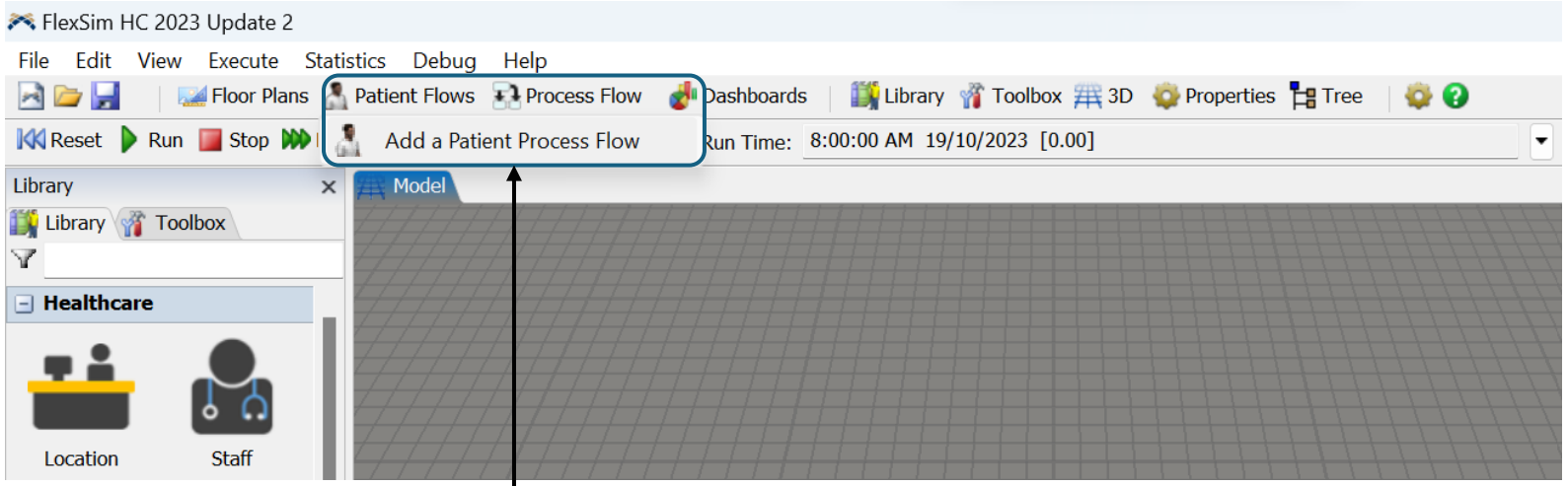
- Recognize the use of HC Activity Sets
- Create Patient Process Flow
- Link 3D model objects to Patient Process Flow
- Key in process data

HC Activity Sets



- Activity sets are a set of preconfigured activities bundled in a container that together model a basic task.
- We can add, remove, and rearrange activities within a set to better suit the task that our model is simulating.

Create Patient Process Flow

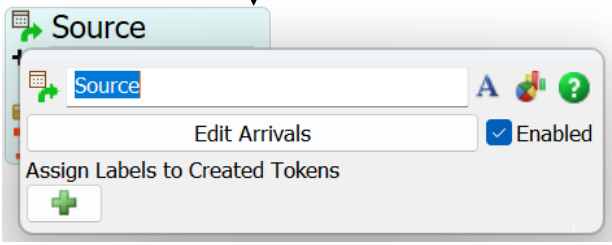


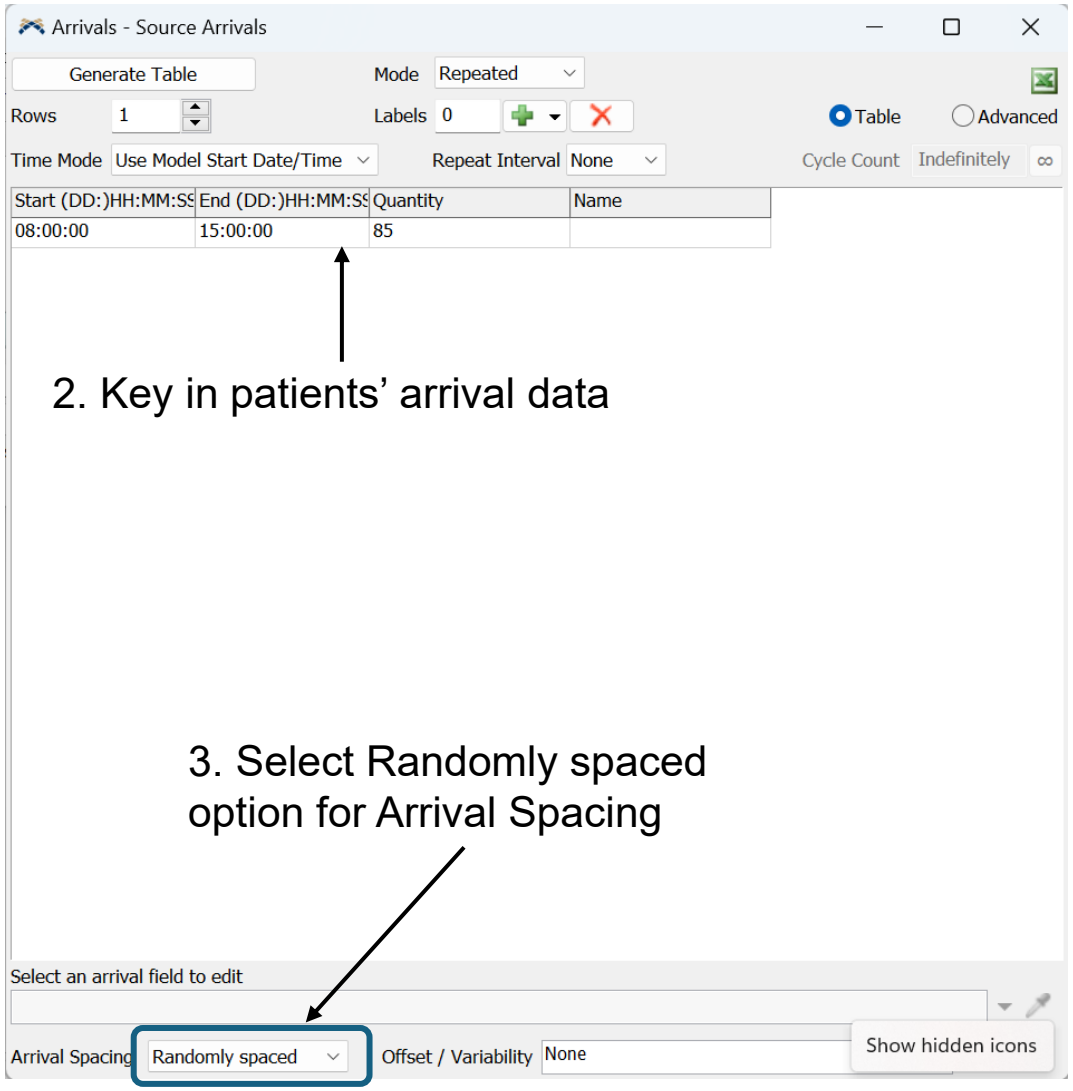
Click Patient Flows and Add a Patient Process Flow.

Two Process Flow windows will be created – Arrivals and Patient Flow

Arrivals – Source

1. Go to Arrivals window. Click on Source and open Edit Arrivals.

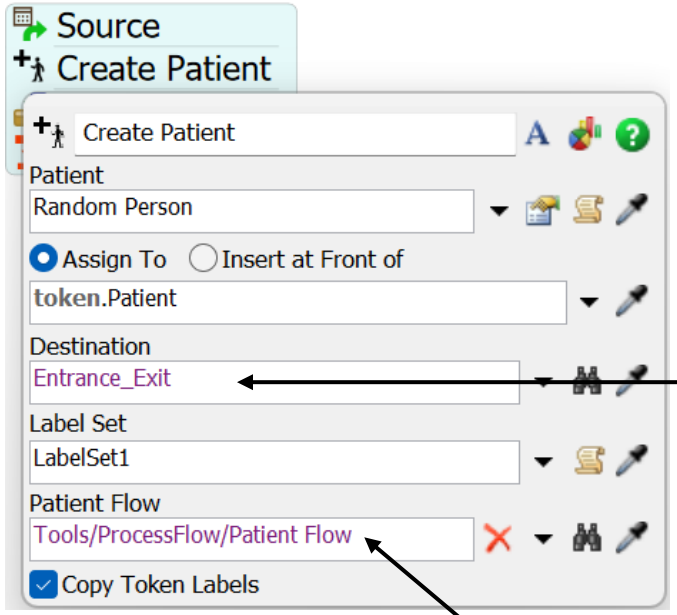




2. Key in patients' arrival data

3. Select Randomly spaced option for Arrival Spacing

Arrivals – Create Patient

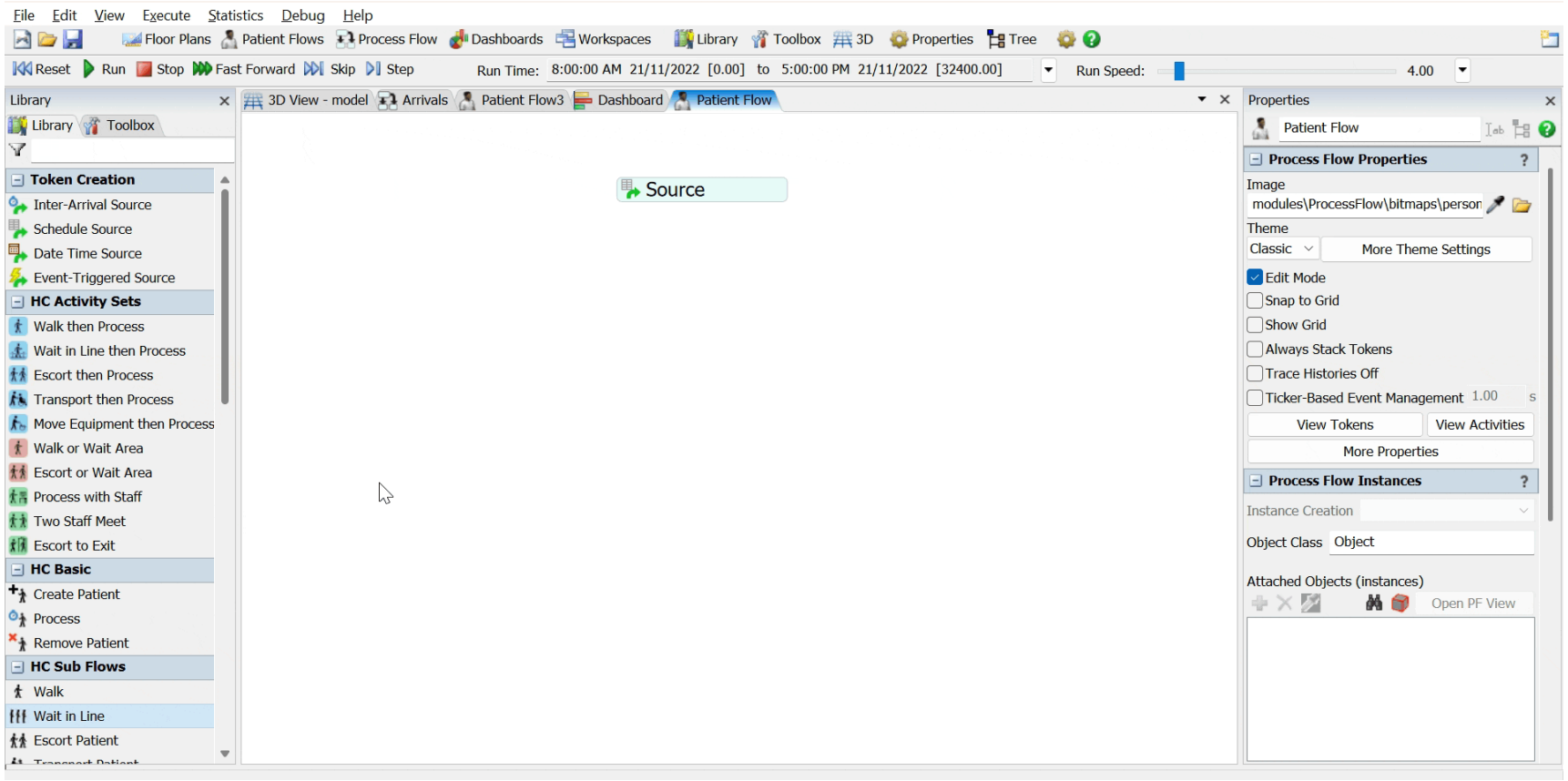


Every patient created will first appear here. Use the sampler tool to select the door prop in your model.

A token will automatically be created in Patient Flow where the activities and control of 3D objects are modeled.

REPUBLIC POLYTECHNIC

Walk then Process Activity Set



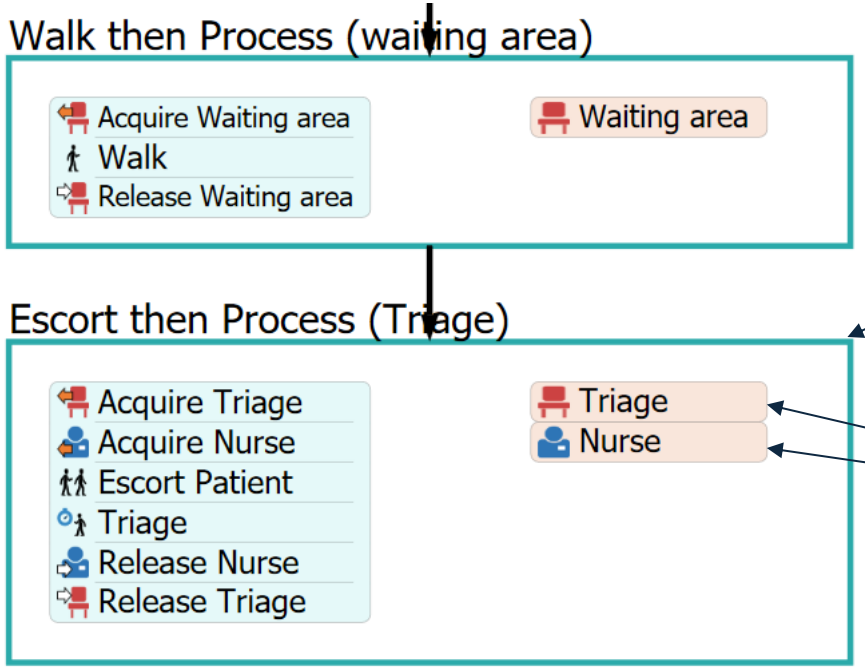
1. Create two Walk then Process Activity Sets.

2. Modify the activity sets as follows:

- Insert **Wait in Line** on top of Acquire Kiosk.
- At Waiting Line input in Wait in Line, use the Sampler to select the waiting line in the 3D model.
- To remove the **Process** after walking to the waiting area, **Hold the Ctrl key,** click & hold the Process activity and drag it out of the stacked block. Delete the Process activity.
- Rename the Location resources and link them to the respective 3D objects.

3. Rename the Activity Sets and link them with the Source using connectors.

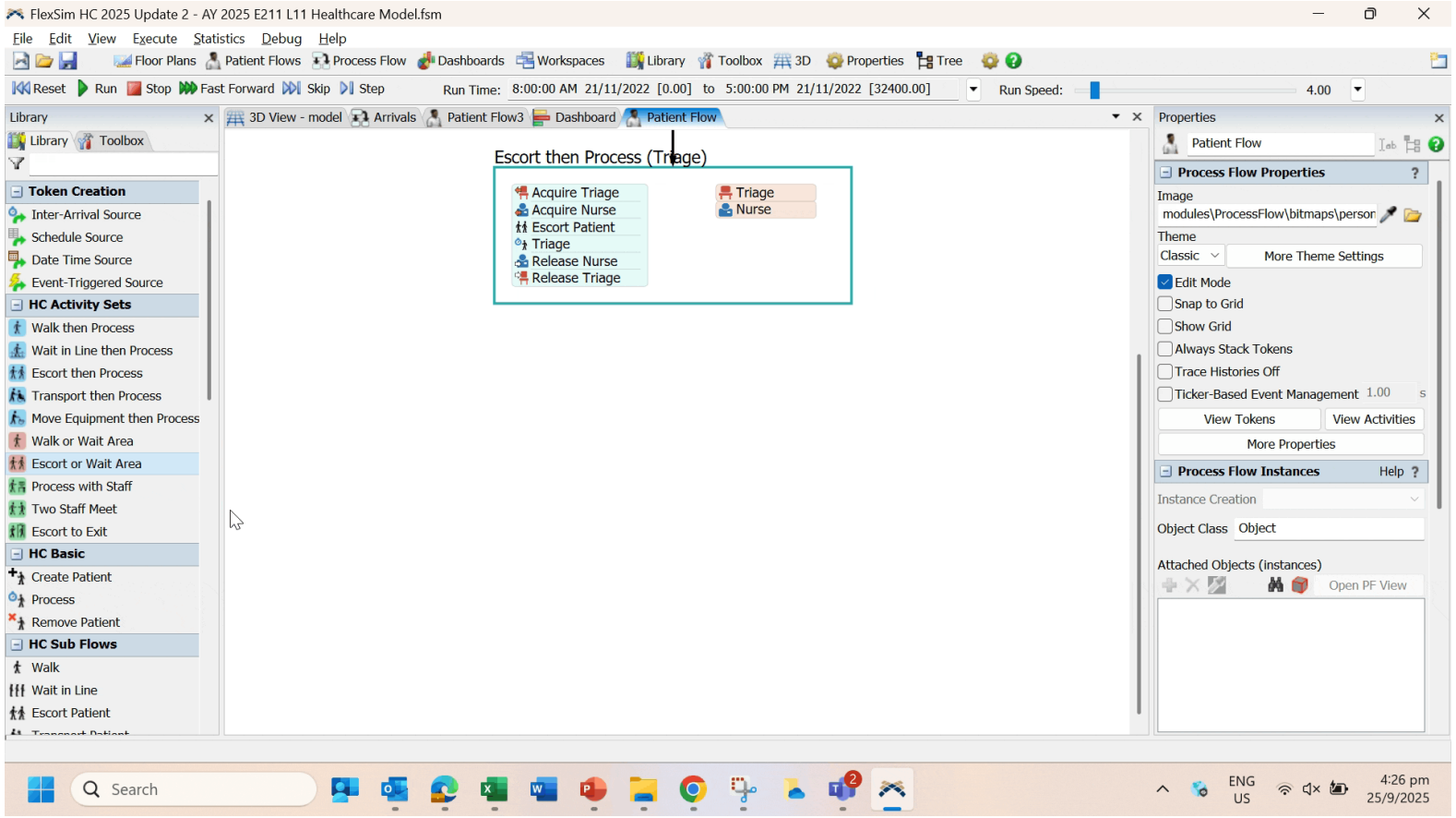
Escort then Process Activity Set



1. Create an Escort then Process Activity Set and connect from Walk then Process.

2. Define the Location & Staff resource
- Link to 3D object
- Rename accordingly

Decide Activity (under Basic)



1. The Decide activity

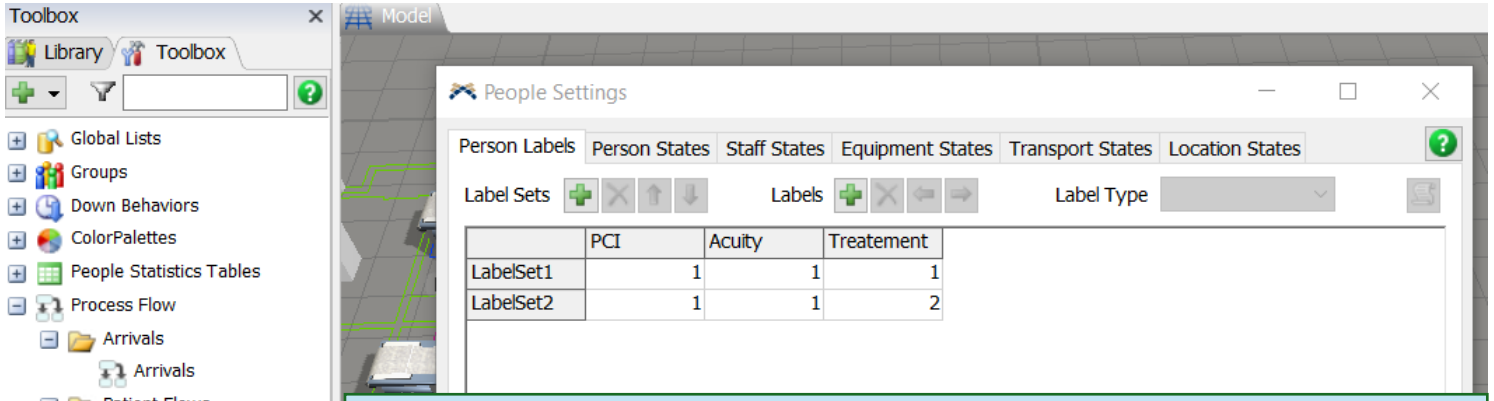
determines which activity a token should be routed to next.

2. Create two activities after Triage:
- Walk then Process (walk back to waiting area to wait for ‘transport’ to Xray)
- Escort or Wait Area (for patient to be escorted to Exam Table)

3. Remember to remove the Process and link the waiting area to its Location resource in Walk then Process

4. Connect them from Decide.

Create Patient Label Set



Create a ‘Treatment’ label set for patients so that the decide activity can direct their flow based on their Treatment label.

- Open the **Toolbox** panel
- Open **People Settings**
- Add a Label Set
- Add a Number label and rename it ‘Treatment’
- Set the value for Treatment to be 1 for LabelSet1 and 2 for LabelSet2.

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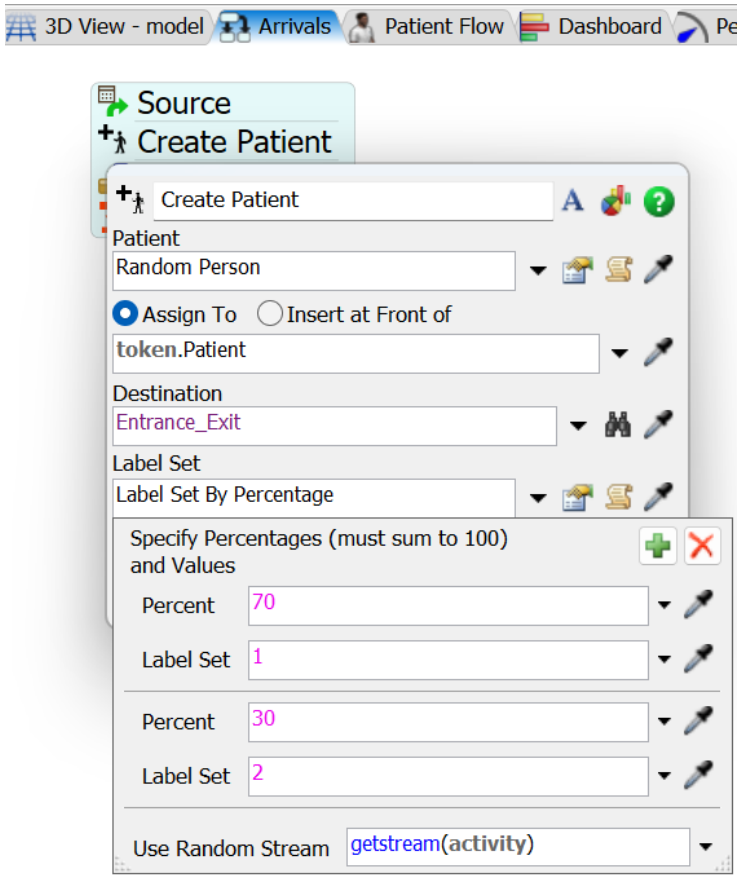
Assign Patients to Different Treatments

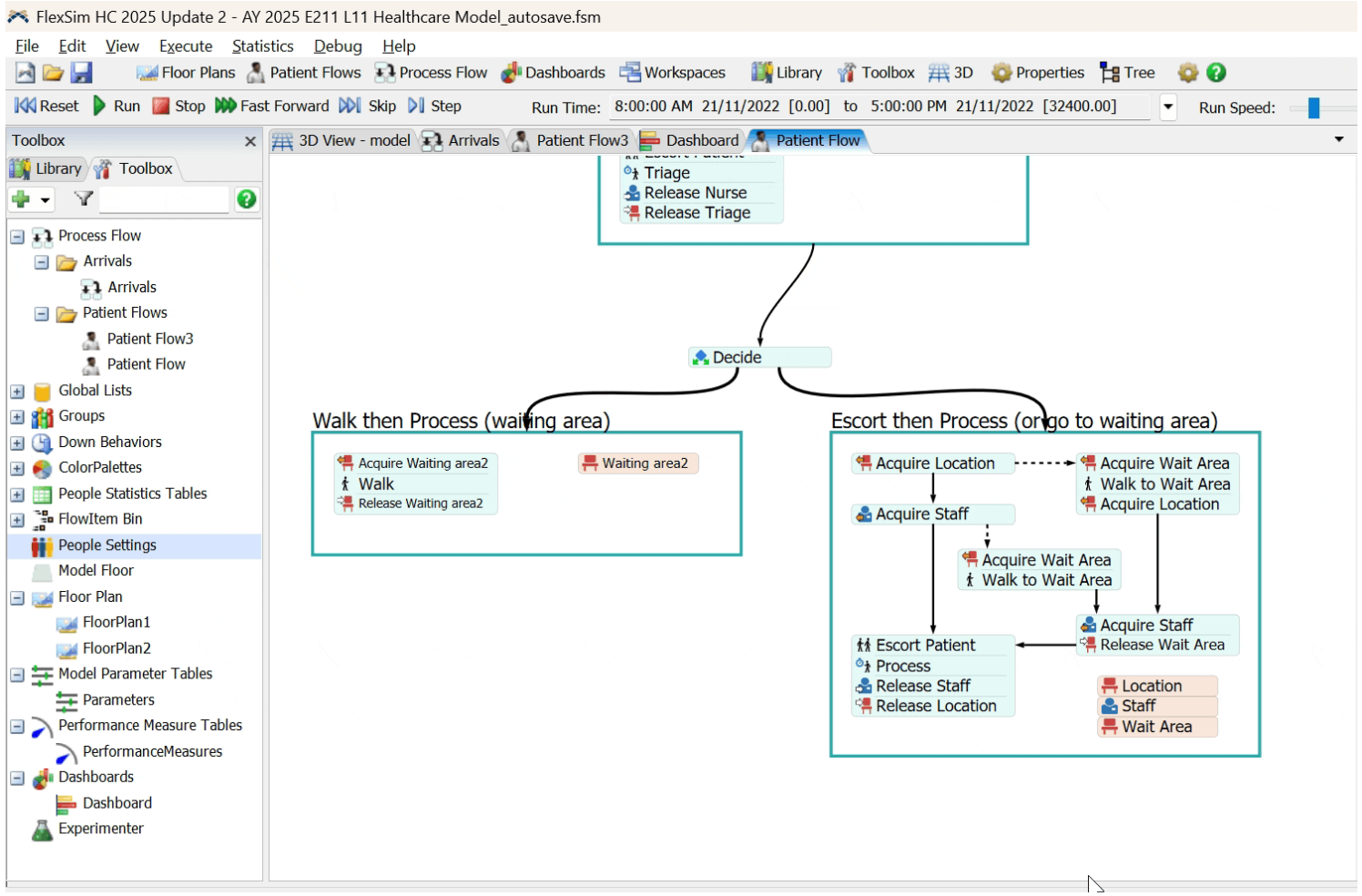


Go to Arrivals.

At the **Create Patient** properties, click the option next to the **Label Set** box to open a menu. Select **Label Set by Percentage**.

Fill out the popup so that 70% of patients are assigned to Label Set 1 and 30% are assigned to Label set 2.

Assign Patients to Different Connector Flow

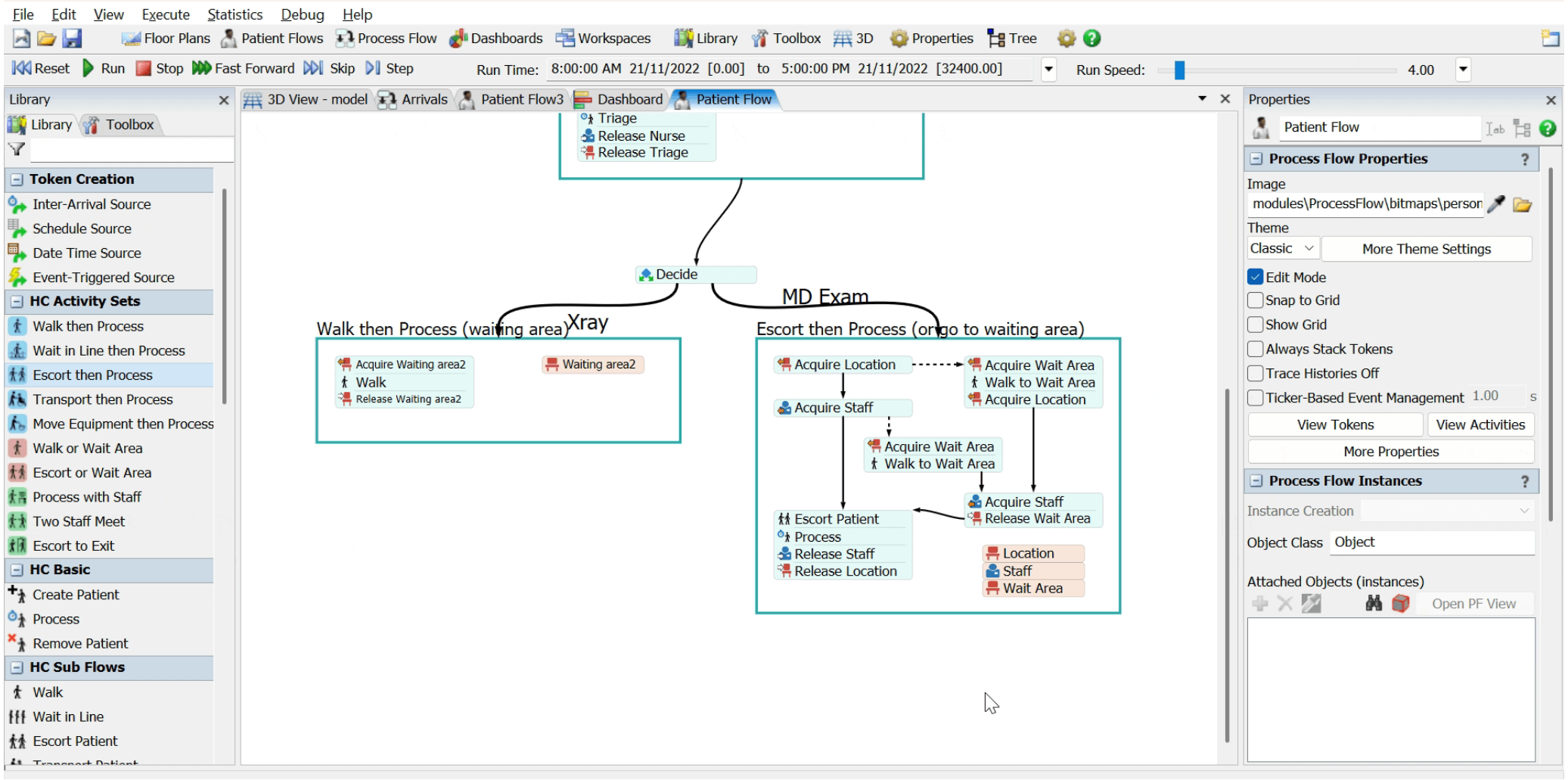


1. At Decide activity, set the Send Token to as patient. Treatment (select Patient Label → Treatment)

2. Since 70% of patients will be examined by MD, make sure connector 1 goes to Escort then Process and connector 2 goes to Walk then Process.

REPUBLIC POLYTECHNIC

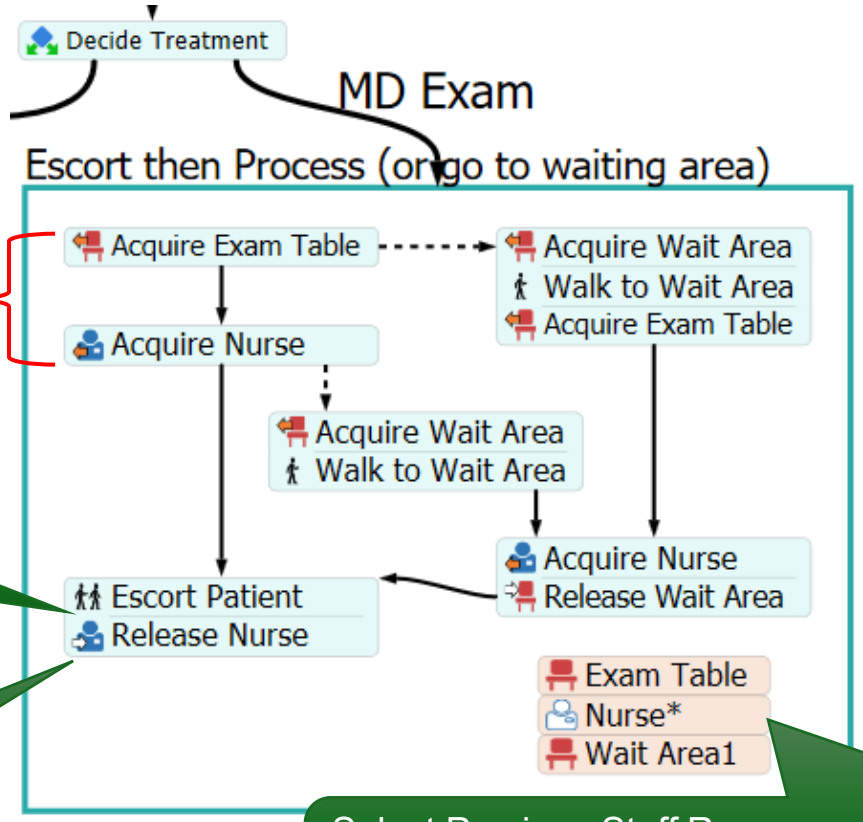
Escort or Wait Area



REPUBLIC POLYTECHNIC

Escort or Wait Area (Explanation)

Two Resources to be acquired – Nurse to provide escort and Exam Table location.
If any of these is not available, Waiting Area will be acquired.



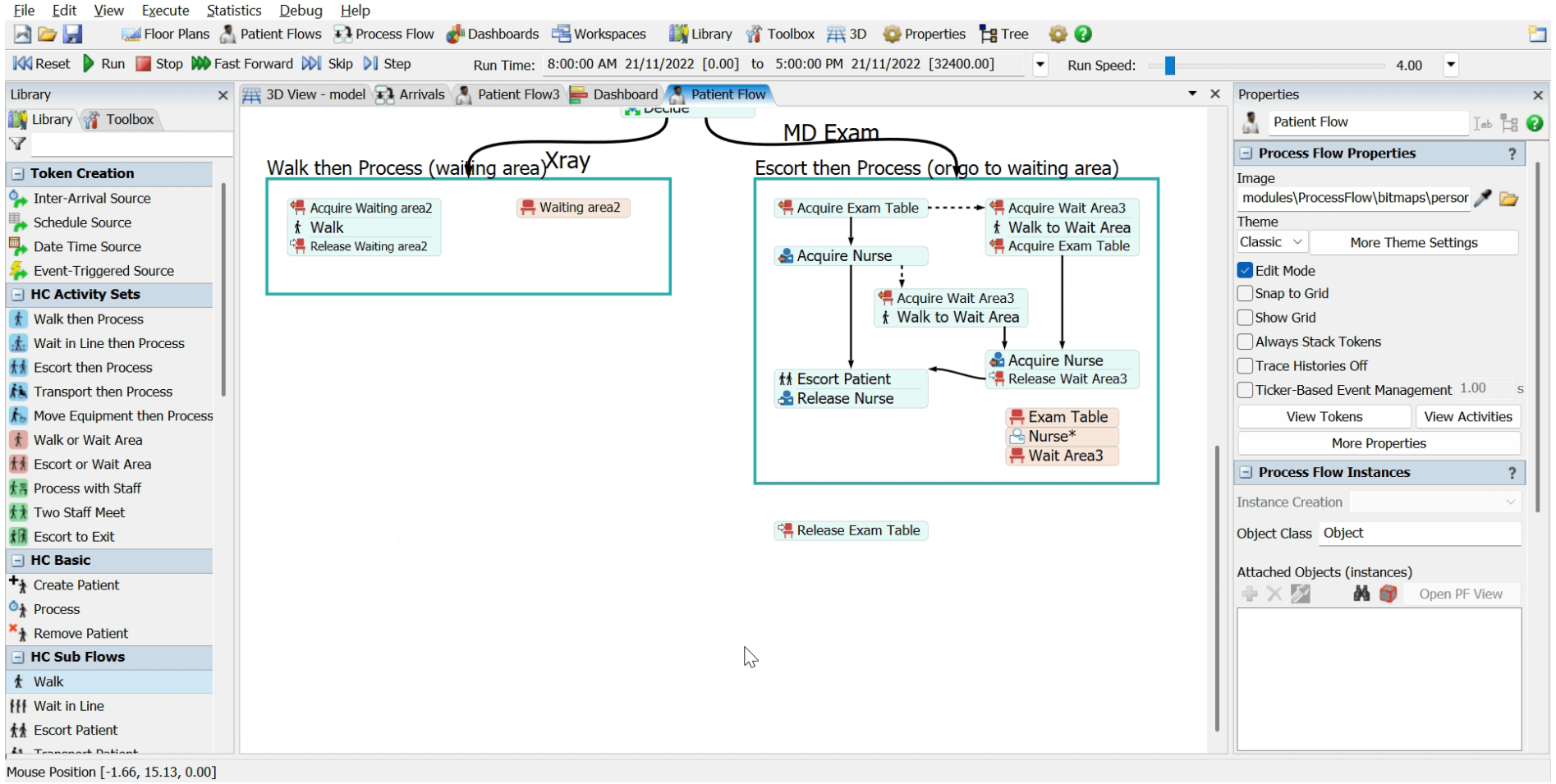
Remove and delete **Process**

Remove but don't delete **Release Location**

Select Previous Staff Resource → Nurse. This will assign the nurse previously attending the patient at Triage to escort the patient.

REPUBLIC POLYTECHNIC

MD Exam path – Process with Staff

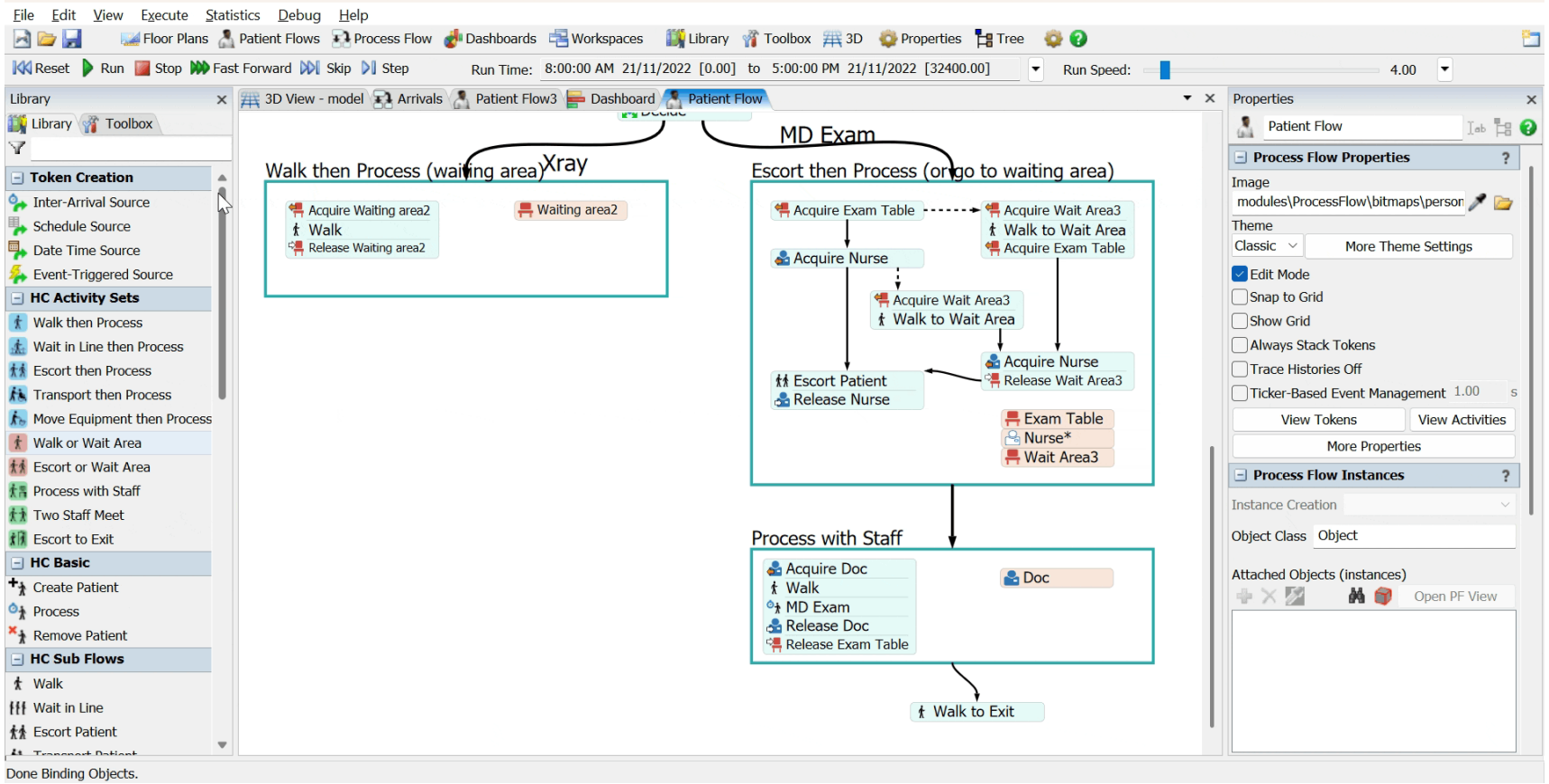


The Exam Table location resource should only be released at the end of MD Exam. Move ‘Release Exam Table’ to after ‘Release Doc’.

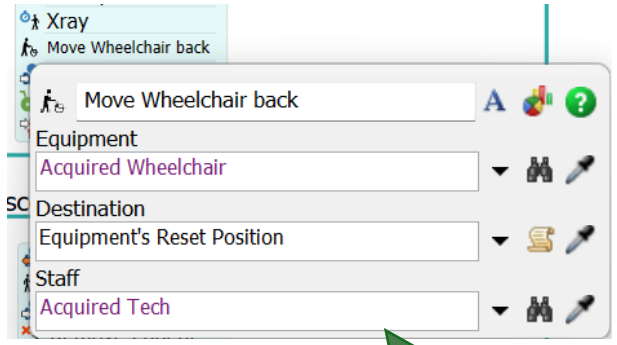
Make sure the activity blocks stay within the Activity box boundary at all times (or else you will get an error during simulation).

REPUBLIC POLYTECHNIC

Xray path – Transport then Process

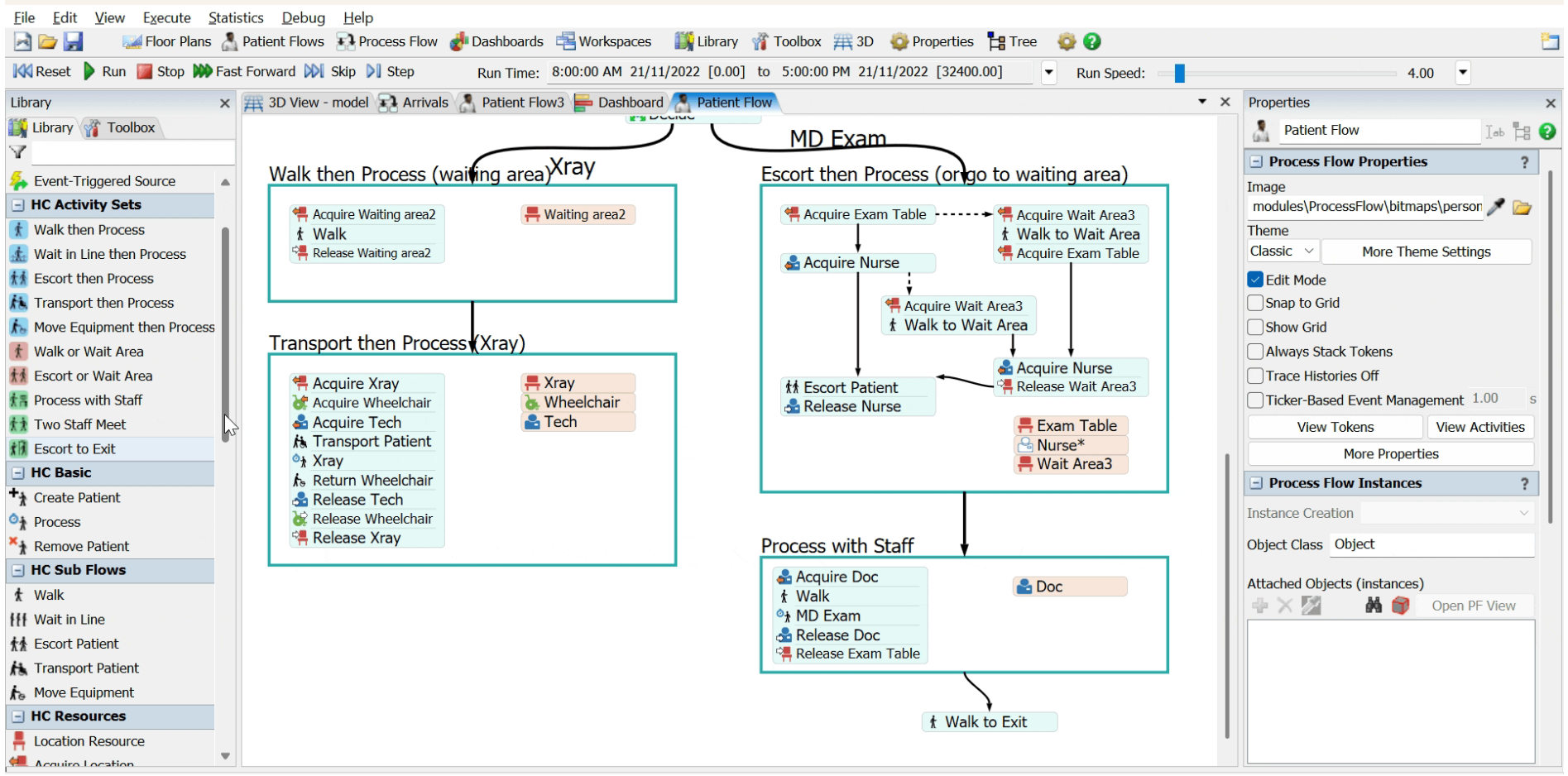


Create a Transport then Process Activity Set.
Insert Move Equipment activity (under HC Sub Flows) after Xray process.



Equipment – Acquired Wheelchair
Destination – Equipment's Reset Position
Staff – Acquired Tech

Xray path – Escort to Exit



Create a Remove Patient activity and connect to the last activities in the two flow paths.

Key in the Process Times and Run Time

Activity ('Process')	Process Time Distribution
Register	Exponential (mean = 2 mins)
Triage	Triangular (min = 2, max = 15, mode = 10 mins)
Xray	Normal (mean = 10 mins, standard deviation = 2 mins)
MD Exam	Exponential (mean = 12 mins)

Run Time: 8:00:00 AM 21/11/2022 [0.00] to 5:00:00 PM 21/11/2022 [32400.00]

Set the Simulation Stop Time at 5:00:00 PM

The End

Upon finishing Part 1 and Part 2 of the model building,
resume your E-Learning in POLITEMall (SCORM page 10).